

Name of Student: _____

Scholar No.: _____

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL**Department of Information Technology****Mid-Term Examination March – 2025**

Subject Name/ Code: - Engg. Mathematics-III (ECE-3001, IT- 3001)

Max. Marks: 20

Time Allotted: 90 min

NOTE: 1. All questions are compulsory.

1.	The pressure and volume of a gas are related by the equation $pv^a = b$, where a and b are constants. Fit this equation to the following set of data:	5														
	<table><tr><td>p(kg/cm²)</td><td>0.5</td><td>1.0</td><td>1.5</td><td>2.0</td><td>2.5</td><td>3.0</td></tr><tr><td>v(litres)</td><td>1.62</td><td>1.00</td><td>0.75</td><td>0.62</td><td>0.52</td><td>0.46</td></tr></table>	p(kg/cm ²)	0.5	1.0	1.5	2.0	2.5	3.0	v(litres)	1.62	1.00	0.75	0.62	0.52	0.46	
p(kg/cm ²)	0.5	1.0	1.5	2.0	2.5	3.0										
v(litres)	1.62	1.00	0.75	0.62	0.52	0.46										
2.	(a) Discuss Similarity and difference between Binomial distribution and Poisson distribution with facts.	2														
	(b) In a book of 300 pages, a proof reader finds no error in 200 pages, in 75 pages one error on each page, in 20 pages two errors on each page and in 5 pages 3 errors on each page. Use Poisson distribution to these data and calculate theoretical frequency. [$e^{-0.43} = 0.6505$]	3														
3.	In a normal distribution, 31% of the items are under 45 and 8% are above 64. Find the mean and standard. Given that if $f(t) = \frac{1}{\sqrt{2\pi}} \int_0^t e^{-x^2/2} dx$ when $f(0.5) = 0.19$, $f(1.4) = 0.42$.	5														
4.	Compute a real root of the equation $10(x - 1) = \sin x$ using the iteration method, correct upto four decimal places.	5														

All the Best